Differentiation: Standard Functions

Questions 1–37

Question 1. Differentiation of $(ax+b)^n$ Differentiate each of the following:

a) $y=(x+3)^5$
b) $y=(4x+1)^6$
c) $y=(6-5x)^4$
d) $y=(2x-3)^7$
e) $y=(7-4x)^{\frac{1}{4}}$
f) $y=\frac{1}{x+2}$
g) $y=\frac{3}{(x-5)^2}$
h) $y=\frac{1}{(8-3x)^4}$
i) $y=\sqrt{2+x}$
j) $y=\frac{1}{\sqrt{3x-1}}$

Question 2. Find $f''(x)$ for each of the following:

a) $f(x)=(7x-1)^5$
b) $f(x)=\sqrt{3-x}$
c) $f(x)=\frac{1}{\sqrt[3]{2x+9}}$

Question 3. Find coordinates of any stationary points on the following curves:

a) $y = (2x - 7)^5$

b) $y = 4x + \frac{1}{x-1}$

Question 4. It is given that $f(x) = (3x + 1)^4$.

a) Find $f'(x)$ and expand the result fully.

b) Expand $f(x)$ using binomial expansion, and hence find $f'(x)$ to confirm your answer to part a).

Question 5. Find the tangent line to the curve $y = \sqrt{x+7}$ when $x = 2$ giving your answer in the form $ax + by = c$ where a, b and c are integers to be found.

Question 6. The height of water, h cm in a harbour for a 12-hour period can be modelled by the equation

$$h = 16(t-6)^2 + 200, \qquad 0 \leq t \leq 12$$

and t is the time in hours after 6 am.

a) By using this model and calculus, predict the minimum height of the water which occurred that day.

b) By considering $\frac{d^2h}{dt^2}$, confirm that your answer to a) is a local minimum.

Question 7. Differentiation of $\sin kx$ **and** $\cos kx$ Differentiate each of the following:

a) $y = \sin 3x$
b) $y = \cos 5x$
c) $y = -7\sin 2x$
d) $y = \frac{1}{2}\cos 6x$
e) $y = -\cos \frac{1}{4}x$
f) $y = -\frac{1}{5}\sin 10x$
g) $y = -\cos \pi x$
h) $y = \cos \frac{x}{\pi}$

Question 8. Find the equation of the tangent to the curve $y = 2\sin 6x$ when $x = \pi$ giving your answer in the form $y = mx + c$.

Question 9. Find the equation for the tangent to the curve $y = 1 + \cos 3x$ when $x = \frac{\pi}{2}$ giving your answer in the form $y = mx + c$.

Question 10. Find the equation for the normal to the curve $y = 3 - \sin(\pi x)$ when $x = 2$ giving your answer in the form $y = mx + c$.

Question 11. Given that $y = \cos 3x$, show that $\frac{d^2y}{dx^2} = -9y$.

Question 12. Show that the curve $y = \cos x - x$ has a single stationary point in the range $0 < x < 2\pi$. Find the exact coordinates of the stationary point.

Question 13. The number of visitors at a tourist attraction, p, may be modelled by the function

$$p = 300 - 40\cos\frac{\pi}{6}t$$

where t is the number of months since 1 January 2024.

a) Find the number of visitors predicted on 1 January.
b) Find an expression for $\frac{dp}{dt}$.
c) Find the value of $\frac{dp}{dt}$ when $t = 6$. What can you deduce in context from this value?
d) Find an expression for $\frac{d^2p}{dt^2}$. What can you deduce from its value when $t = 6$?

Question 14. The distance, s cm, of a weight dropped from an equilibrium position may be modelled by the equation

$$s = 3\cos\frac{\pi}{3}t$$

where t is the number of seconds after the weight is released.

a) Find an expression for $\frac{ds}{dt}$.
b) Find the maximum value of $\frac{ds}{dt}$ giving your answer in exact form and find the first time when this occurs. Explain the relevance of this point in context.

Question 15. Given the gradient of the normal to the curve $y = 6\cos 2\pi x$, $0 < x < \frac{\pi}{3}$ is $\frac{1}{6\pi}$, find the possible values of x for when this occurs giving your answer in exact form.

Question 16. Differentiate $y = \sin x^\circ$, giving your answer in degrees.

Question 17. Differentiation of e^{kx}, $\ln kx$, a^x Differentiate each of the following:

a) $y = \ln 5x$
b) $y = 3^x$
c) $y = e^{5x}$
d) $y = 2\ln \frac{1}{4}x$
e) $y = -6^x$
f) $y = (e^x - 1)^2$

Question 18. For each curve, find the coordinates of the stationary points and identify their type:

a) $y = 2e^x - 4x$
b) $y = \ln x - 8x$
c) $y = 2x^2 + 19x - 5\ln x$

Question 19. Find the x-value for which $f'(x) = 1$ giving your answers in exact form:

a) $f(x) = \ln 2x$

b) $f(x) = e^{2x} + 4$

c) $f(x) = 3^x - 5$

Question 20. Prerequisite knowledge for Q20: Points of Inflection Show that the curve $y = 8\ln x - 5x^2$ is concave (concave downwards) for all values of $x > 0$.

Question 21. Find the equation of the tangent to the curve $y = \ln 5x$ when $x = \frac{e}{5}$ giving your answer in the form $y = kx$ where k is a constant.

Question 22. Find the equation of the normal to the curve $y = 5^x$ when $x = 2$ in the form $ay + bx = c$ where a, b and c are constants given in exact form.

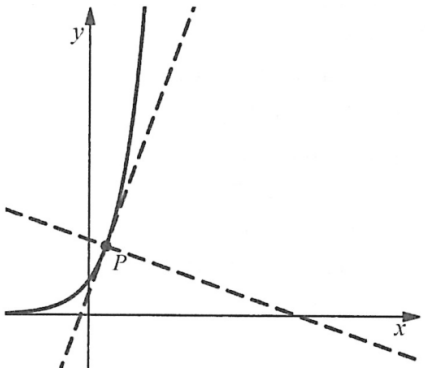
Question 23. A population P of bugs has been found and a treatment applied so their population is reduced. The number of bugs after t days can be modelled by

$$P = 500 \times 20^{-\frac{1}{2}t}$$

a) Determine how many whole hours need to have elapsed for the number of bugs to drop to 50.
b) Determine the rate of change of bugs after 7 days.

Question 24. Find the coordinates of the single stationary point of the curve $y = 9^x - 3^x$ and determine its type.

Question 25. The graph below shows the curve with equation $y = 2 \times 2^x$.



The point P lies on the curve at $x = 1$. By finding the equation of the tangent and normal line at the point P, find the area of the triangle bounded by these two lines and the x-axis. Give your answer in the form

$$p \times \frac{1}{\ln 2} + q \ln 2$$

where p and q are constants to be found.

Question 26. Show that the equation of the tangent to the curve $y = \ln x + a$ when $x = e^b$ is

$$e^b y - x = (a + b - 1)e^b$$

for constants a and b.

Question 27. Differentiation of $\tan kx$ Differentiate each of the following:

a) $y = \tan 3x$

b) $y = 2\tan 4x$

c) $y = -\frac{1}{3}\tan 9x$

d) $y = -\tan \pi x$

e) $y = -\frac{1}{12}\tan \frac{6}{\pi}x$

Question 28. Find the equation for the tangent to the curve $y = 4 + 2\tan \frac{\pi}{9}x$ when $x = \frac{9}{4}$ in the form $y = mx + c$.

Question 29. Show that the equation for the normal to the curve $y = x - 2\tan\frac{\pi}{4}x$ when $x = 4$ crosses the x-axis at $x = 8 - 2\pi$.

Question 30. Mix of Standard Functions Find the gradient function of each of the following:

a) $y = 3\sin 7x + 2\tan 2x$
b) $y = 15^x - 3\cos\frac{5}{2}x$
c) $y = \ln 180x + \tan\sqrt{2}x$
d) $y = \sqrt{x} - 3\sin 2x$
e) $y = e^{3x} + \sin 8x$
f) $y = \cos\frac{1}{4}x - \cos 2\pi x$
g) $y = 4^x + \tan \pi x$

Question 31. Find $\frac{d^2y}{dx^2}$ for each of the following:

a) $y = 3x^5 - e^{5x}$
b) $y = x^2 - \sin 2x$
c) $y = \sqrt{x} + \ln 2x$
d) $y = \frac{1}{x} - \cos \frac{1}{2}x$
e) $y = e^{2x} - \sin \pi x$
f) $y = e^{-\frac{1}{2}x} + \sin 3x - \cos 3x$
g) $y = \frac{1}{\sqrt{x}} - \ln \frac{1}{2}x$

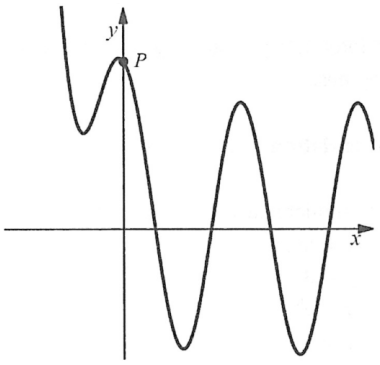
Question 32. Find the exact value of $f'(x)$ at the given value of x, giving your answer as an exact value:

a) $f(x) = e^{2x} + x^{\frac{1}{4}}$ when $x = 1$
b) $f(x) = 5 \ln x + \frac{1}{x^2}$ when $x = 2$
c) $f(x) = 2^x + \sin 2\pi x$ when $x = \frac{1}{2}$
d) $f(x) = \cos(\frac{\pi}{2}x) + \tan(-\frac{\pi}{2}x)$ when $x = 0$

Question 33. Determine whether the curve $y = e^{-2x} - 3\sin 5x$ is increasing or decreasing when $x = \pi$.

Question 34. Find the equation for the tangent to the curve $y = \ln 2x + \sin \pi x$, $x > 0$ in the form $y = mx + c$ when $x = 3$.

Question 35. The curve shown below is $y = e^{-2x} + 3\cos \pi x$.



Point P is where the curve meets the y-axis.

a) Find the tangent line to the curve at P.
b) Find the normal line to the curve at P.
c) Find the area bounded by the normal and tangent lines at point P and the x-axis.

Question 36. Prerequisite knowledge for Q36: Points of Inflection Show that the curve $y = 2^x + \cos 3x$

a) is convex (concave upwards) at $x = \pi$
b) is concave (concave downwards) at $x = \frac{2\pi}{3}$

Question 37. Show clearly that the curve $y = \sin 3x + \tan 3x$ has a single stationary point for $0 < x < \pi$ and state the coordinate of this stationary point.